WORKED SOLUTION

STEP II 2006, Question 7

Pure • Coordinate geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q7

The key idea

The tangent comes quickest from the "replace x^2 by xx_0 and y^2 by yy_0" form, or equally from implicit differentiation. For the construction, just work out the coordinates of Q and the equation of PQ, then compare with the tangent at a general point. The words $\tan(\alpha/2) = k$ are a strong hint: they tell you to write $\cot\alpha$ and $\operatorname{cosec}\alpha$ in terms of $t = \tan(\alpha/2)$.

Solution

The tangent

Differentiating $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ implicitly with respect to x gives

$$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 0, \qquad \text{so} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{b^2x}{a^2y}.$$

At $(a\cos\alpha, b\sin\alpha)$, with $\sin\alpha \neq 0$ (otherwise the tangent is vertical and the stated equation makes no sense), the gradient is

$$-\frac{b^2 a\cos\alpha}{a^2 b\sin\alpha} = -\frac{b\cot\alpha}{a}.$$

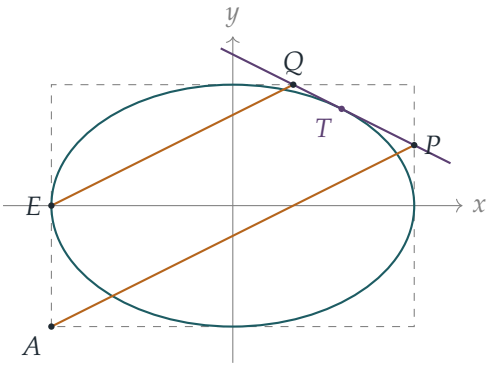
So the tangent is

$$\begin{aligned} y - b\sin\alpha &= -\frac{b\cot\alpha}{a}(x - a\cos\alpha) \\ \implies \quad y &= -\frac{b\cot\alpha}{a}x + b\frac{\cos^2\alpha}{\sin\alpha} + b\sin\alpha = -\frac{b\cot\alpha}{a}x + \frac{b(\cos^2\alpha + \sin^2\alpha)}{\sin\alpha}, \end{aligned}$$

that is, $y = -\dfrac{b\cot\alpha}{a}x + b\operatorname{cosec}\alpha$, as required.

The construction

Here is the picture with $a = 3$, $b = 2$, $k = \frac{1}{2}$. The ellipse sits inside the rectangle with corners $(\pm a, \pm b)$; A is the bottom-left corner, E the midpoint of the left side, P is on the right side and Q on the top side.



The line AP joins $(-a, -b)$ to (a, kb), so its gradient is

$$\frac{kb - (-b)}{a - (-a)} = \frac{(1+k)b}{2a}.$$

The parallel line through $E(-a, 0)$ is $y = \dfrac{(1+k)b}{2a}(x + a)$. Putting $y = b$ gives $x + a = \dfrac{2a}{1+k}$, so

$$Q = \left(\frac{a(1-k)}{1+k},\ b\right).$$

Now the gradient of PQ is

$$\frac{b - kb}{\dfrac{a(1-k)}{1+k} - a} = \frac{b(1-k)}{\dfrac{-2ak}{1+k}} = -\frac{b(1-k^2)}{2ak},$$

which is fine because $k \neq 0$. Using the point $P(a, kb)$, the line PQ is

$$y = kb - \frac{b(1-k^2)}{2ak}(x - a) = -\frac{b(1-k^2)}{2ak}\,x + \frac{2k^2 b + b(1-k^2)}{2k},$$

so

$$PQ: \qquad y = -\frac{b}{a} \cdot \frac{1-k^2}{2k}\,x + b \cdot \frac{1+k^2}{2k}.$$

Ask yourself How do we recognise this as the tangent at some point of the ellipse?

Compare it with the tangent at $(a\cos\alpha, b\sin\alpha)$ found above. We need an α with $\cot\alpha = \dfrac{1-k^2}{2k}$ and $\operatorname{cosec}\alpha = \dfrac{1+k^2}{2k}$. These are exactly the t-formulae: with $t = \tan(\alpha/2)$,

$$\tan\alpha = \frac{2t}{1-t^2}, \qquad \sin\alpha = \frac{2t}{1+t^2}, \qquad \text{so} \qquad \cot\alpha = \frac{1-t^2}{2t}, \qquad \operatorname{cosec}\alpha = \frac{1+t^2}{2t}.$$

Since $0 < k < 1$, we can choose $\alpha = 2\arctan k$, which lies in $(0, \frac{\pi}{2})$, so $\sin\alpha \neq 0$ and $\tan(\alpha/2) = k$. With $t = k$ the two expressions match both the gradient and the intercept of PQ. So PQ is the line $y = -\frac{b\cot\alpha}{a}x + b\operatorname{cosec}\alpha$, which is the tangent to the ellipse at $(a\cos\alpha, b\sin\alpha)$ where $\tan(\alpha/2) = k$. This is what we had to show.

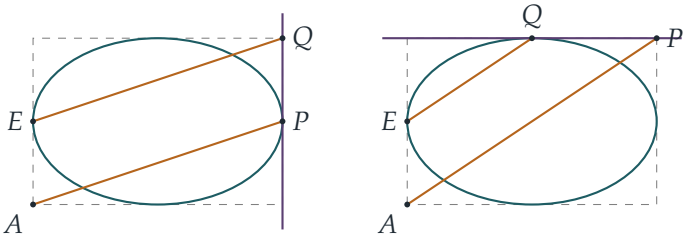
In coordinates, the point of contact is $T = \left(\dfrac{a(1-k^2)}{1+k^2}, \dfrac{2kb}{1+k^2}\right)$.

The cases $k = 0$ and $k = 1$

The algebra above divided by k, and the tangent formula needs $\sin\alpha \neq 0$, so we redo the construction directly.

$k = 0$. Now $P = (a, 0)$, and AP has gradient $\frac{b}{2a}$. The parallel line through E is $y = \frac{b}{2a}(x + a)$, which meets $y = b$ at $x = a$, so $Q = (a, b)$. Then PQ is the vertical line $x = a$. That is the tangent to the ellipse at $(a, 0)$, the point with $\alpha = 0$, and $\tan(0/2) = 0 = k$. So the result *does* still hold, even though the tangent can't be written in the form $y = mx + c$.

$k = 1$. Now $P = (a, b)$, and AP is the diagonal of the rectangle, with gradient $\frac{b}{a}$. The parallel line through E is $y = \frac{b}{a}(x + a)$, which meets $y = b$ at $x = 0$, so $Q = (0, b)$. Then PQ is the horizontal line $y = b$, the tangent at the top of the ellipse $(0, b)$. This is the point with $\alpha = \frac{\pi}{2}$, and $\tan(\pi/4) = 1 = k$. So the result holds here too (and the formula works: $\cot\frac{\pi}{2} = 0$ and $\operatorname{cosec}\frac{\pi}{2} = 1$ give $y = b$).



Checks

- Take $a = 3$, $b = 2$, $k = \frac{1}{2}$. Then $Q = (1, 2)$ and PQ is $y = -\frac{1}{2}x + \frac{5}{2}$. Also $\tan\alpha = \frac{4}{3}$, so $\cos\alpha = \frac{3}{5}$, $\sin\alpha = \frac{4}{5}$ and $T = (\frac{9}{5}, \frac{8}{5})$. This lies on PQ ($-\frac{9}{10} + \frac{5}{2} = \frac{8}{5}$), and the gradient there is $-\frac{b^2x}{a^2y} = -\frac{4\cdot 9/5}{9\cdot 8/5} = -\frac{1}{2}$.
- Substituting PQ into the ellipse equation should give a quadratic with a repeated

root. We checked this, and the formulae for Q and PQ, numerically for a thousand random a, b, k.

- As $k \to 0^+$, $T \to (a, 0)$, and as $k \to 1^-$, $T \to (0, b)$. This agrees with the two end cases.

Common slips

- Getting the sign wrong in the gradient of PQ: the denominator $\frac{a(1-k)}{1+k} - a$ is negative.
- Saying the result fails at $k = 0$ because $\operatorname{cosec} 0$ is undefined. The *formula* breaks down, but the geometry doesn't: PQ is still the tangent at $\alpha = 0$.

Beyond STEP

The substitution $t = \tan(\alpha/2)$ is a rational parametrisation of the ellipse: $\left(a\frac{1-t^2}{1+t^2}, b\frac{2t}{1+t^2}\right)$. It is the same idea that turns trigonometric integrals into rational ones and finds all Pythagorean triples. Every conic can be parametrised rationally like this, and constructions like the one here, built only from parallels and intersections, are the subject of projective geometry. For a first look at how this connects with number theory, try Davenport, *The Higher Arithmetic*.

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